

CHEN, CIN-YU

sarah871230@gmail.com , (+886)0937085376

EDUCATION	M.Sc. in Mathematical Sciences 2022–2024 National Chengchi University, Taiwan GPA: 4.24 / 4.3
	B.Sc. in Statistical Science 2017–2022 Department of Applied Mathematics, National Dong Hwa University, Taiwan GPA: 3.98 / 4.5
RESEARCH INTERESTS	Stochastic analysis, martingale theory, mathematical finance, pricing of contingent claims, and optimal stopping.
RESEARCH EXPERIENCE	<i>Master's Thesis:</i> 2024 <i>National Chengchi University</i> A Study of Discrete-Time Martingales with Applications to the Kalman Filter, ◦ Focusing primarily on discrete-time martingale theory ◦ With a structured introduction on the Kalman filter and its interpretation under partial information control problems.
SEMINARS	Probability & Martingales 2022–2023 <i>(M.Sc. Year 1)</i> <i>National Chengchi University</i> ◦ Graduate seminar presentations on discrete-time martingales, stopping times, and measure-theoretic foundations of probability. ◦ Based on R. Durrett, <i>Probability: Theory and Examples</i>.
	Stochastic Calculus 2023–2024 <i>(M.Sc. Year 2)</i> <i>National Chengchi University</i> ◦ Graduate seminar presentations on Brownian motion, continuous-time martingales, and stochastic integration. ◦ Based on J.-F. Le Gall, <i>Brownian Motion, Martingales, and Stochastic Calculus</i>.
	Mathematical Finance 2024– present <i>(Research assistant)</i> <i>National Central University</i> ◦ Seminar-style expository presentations on discrete-time financial models and stochastic calculus for finance. ◦ Based on S.E. Shreve, <i>Stochastic Calculus for Finance</i> and R.J. Williams, <i>Introduction to the Mathematics of Finance</i>.
RESEARCH NOTES	Research notes developed along with RA seminar presentations. • Pricing European Claims via Replication and Martingale Properties ◦ Replication-based pricing of European claims and its connection to martingale price processes. • Pricing American Contingent Claims via Superhedging

- Seller-based superhedging price and the dual interpretation via optimal stopping.

- **FTAP I and II: Interpretation and Implications for Pricing**

- Conceptual study of the fundamental theorems of asset pricing.

COURSEWORK
(*Graduate Level*)

Probability and Measure Theory
Stochastic Processes
Stochastic Calculus
Real Analysis

**SHORT
COURSES**

An Introduction to Diffusion Processes,
National Center for Theoretical Sciences (NCTS) , Taiwan Oct 2025 (3 days)

A Brief Discussion on Brownian Motion and Related Processes with Applications,
National Center for Theoretical Sciences (NCTS) , Taiwan July 2023 (2 weeks)

**HONORS &
AWARDS**

Honorary Member, Phi Tau Phi Scholastic Honor Society Graduate 2024
Academic Excellence Award (Dean's List), multiple semesters Undergraduate

SKILLS

LaTeX

**TEACHING
EXPERIENCE**

Teaching Assistant, Calculus, National Chengchi University 2023-2024